Analytical Cumulative Examination

4/21/06

1. (35 pts) The protein metmyoglobin, and the azide ion (N_3^-) (nonabsorbing) form a complex. The molar absorption coefficients ($M^{-1}cm^{-1}$) for metmyoglobin (Mb) and the complex (C) in a buffer are given below:

λ (nm)	$\varepsilon_{Mb} \times 10^{-4}$	$\varepsilon_C \times 10^{-4}$
490	0.850	0.744
540	0.586	1.028

A mixture of Mb and N_3^- in the same 1.00 cm cell gave an absorbance of 0.656 at 490 nm, and 0.716 at 540 nm.

a. What are the concentrations of metmyoglobin and the complex in the mixture?
b. If the total azide concentration is 1.05×10^{-4} M, what is the equilibrium constant for the formation of the complex.

$$Mb + N_3^- \Longleftrightarrow C$$

2. (35 pts) Since the energy of a molecular quantum state is divided by kT in the Boltzmann equation, it is of interest to determine the temperature at which kT is equal to the energy of photons of various wavelengths. Calculate the temperature at which kT is equal to the energy of a photon in the microwave region, the infrared, and the sodium D line (590 nm).

3. (30 pts) The specific rotation of l-leucine is $[\alpha] = -14.0$ deg. (cm g/mL)$^{-1}$. If the specific rotation of a mixture of l and d forms is +2.3 deg, calculate the fraction of l form in the mixture.